

Process modeling. Basic principles.



Why modeling is necessary?

- Business characteristics
 - Globalization
 - Geographical dispersion
 - Drive for management and operational efficiency
- IT Specifics
 - Wide range of platforms
 - Integration

Why modeling is necessary?

Business modeling makes it possible to determine not only how the company operates as a whole and interacts with external organizations, customers, and suppliers, but also how activities are organized at each individual workstation.

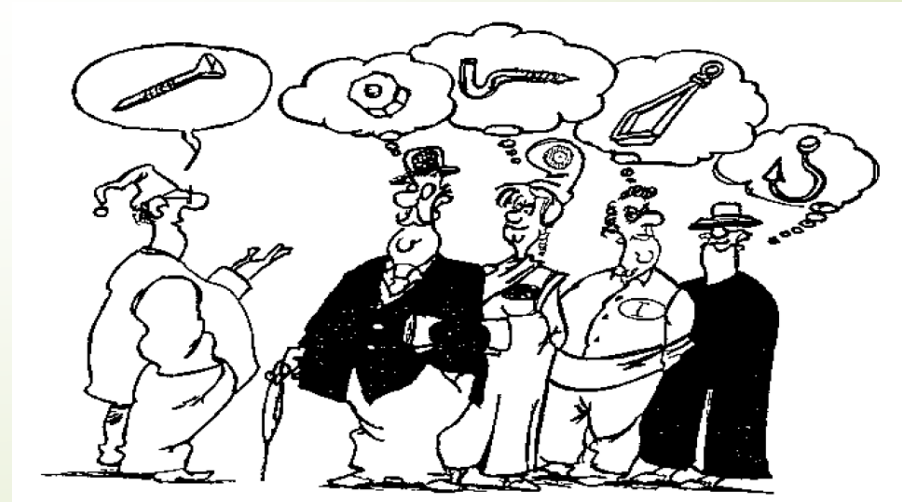
A model is a set of objects and the relationships between them that adequately describes only certain properties of the system being modeled.

The model is merely one of many possible interpretations of the system. This interpretation must be satisfactory to the user in the given situation and at the given moment in time.



What is the purpose of a model?

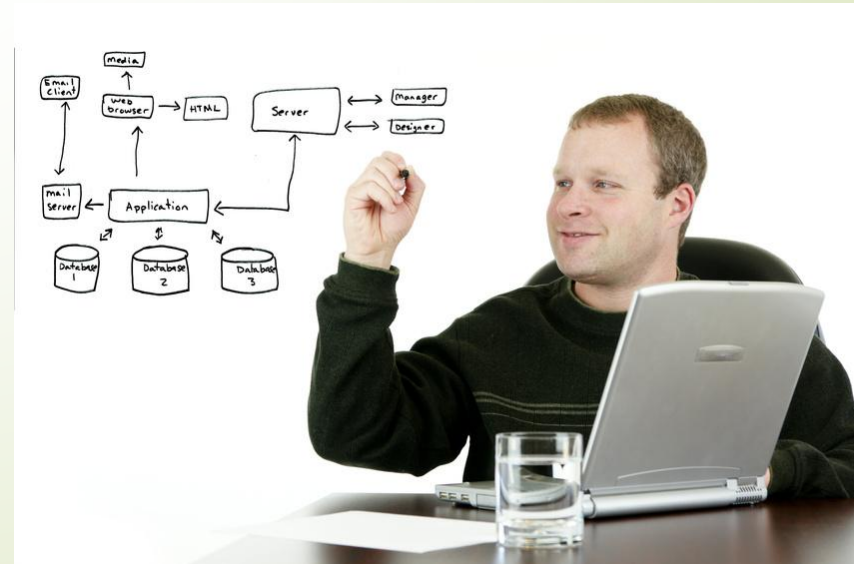
- Show relationships between objects
- *Make the system description easier to grasp*
- Focus attention on a certain aspect of the system
- *Describe the system using standard notation that precludes ambiguous interpretation.*



Business model

A business model, like any model, is a simplified representation of a real-world object (a business system).

In tangible terms, it is a system of graphic and textual descriptions that enable an understanding of the essence of the enterprise management process.



What is a business model used for?

- Business models serve as a link between the IT implementation of the system and the business.
- *Business models are a key element of information system development.*
- Business models make it possible to analyze various options for changing business behavior without implementing them in practice.

The methodology (or notation) for creating a business process model (or description) is understood as the set of methods used to represent real-world objects and the relationships between them in the form of a model.

Any methodology (or method) includes:

- the theoretical basis;
- a description of the steps required to achieve the specified result;
- recommendations for use, both individually and as part of a set of methods.

The graphical model notation comprises:

- a strictly defined set of interconnected graphical representations — elements of the graphical language;
- various types of connections between them;
- text fragments (natural language);
- embedded objects;
- a glossary.

- SADT: IDEF0, IDEF3, DFD
- ARIS methodology: VAD, eEPC
- RUP: UML, Use cases, Activity diagrams
- BPM: BPMN, BPEL

Stages of the simulated system's life cycle:

- data collection and hypothesis formulation;
- design of the model (and sub-model) structure and composition;
- development and debugging of sub-models, and assembly of the complete model;
- investigation of the model;
- assessment of model adequacy;
- analysis of simulation results;
- model refinement and modification.



Modeling rules:

- identifying the set of models to be used, interpreting the selected models, and establishing rules for their construction;
- structuring information when creating an organizational model;
- determining the level of modeling detail;
- defining naming conventions for objects and models;
- defining rules for the semantic validation of the created models.

When selecting a set of models, the following should be considered:

- *The set of models depends directly on the modeling objectives and the expected outcome.*
- *Each model presents only a limited view of the subject area.*
- *Models can complement each other.*
- *Models can serve as alternatives to one another.*

Modeling Agreement:

- List of terms and abbreviations
- Basic document formatting principles
- General principles of model development
- Document templates



Types of modeling methodologies

The following methodologies are among the most common types:

- Business Process Modeling
- *Work Flow Modeling*
- Data Flow Modeling



The key system-forming elements are:

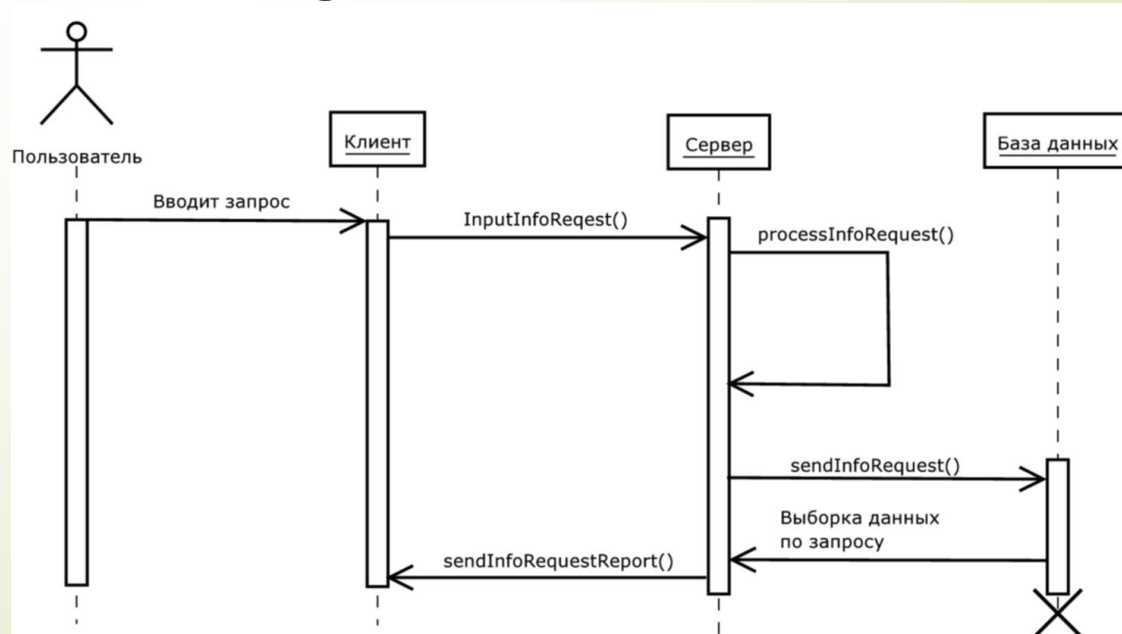
- the company's organizational structure;
- the territorial structure;
- the composition and structure of the company's core business processes;
- the classification and structure of key working documents;
- the classification and structure of information systems.

Key stages:

- construction of a hierarchy of goals for optimizing organizational activities
- *selection of methodology*
- selection of models
- *analysis of the organization's activities*
- development of models in accordance with the hierarchy of goals
- *model optimization*
- reorganization of operations

Detailed modeling

- ▶ a set of use cases reflecting possible business process execution scenarios
- ▶ activity diagrams detailing the sequence of business process execution
- ▶ interaction diagrams illustrating document workflow schemes



Why model systems?

Modeling makes it possible to accomplish four tasks:

- visualize the system in its current or desired state
- define the system's structure or behavior
- obtain a template for constructing the system
- document decisions made using the resulting models

Complex systems must be modeled because, otherwise, it is impossible to perceive them as a unified whole.

UML (Unified Modeling Language) is a language for visualizing, specifying, constructing, and documenting software system artifacts.

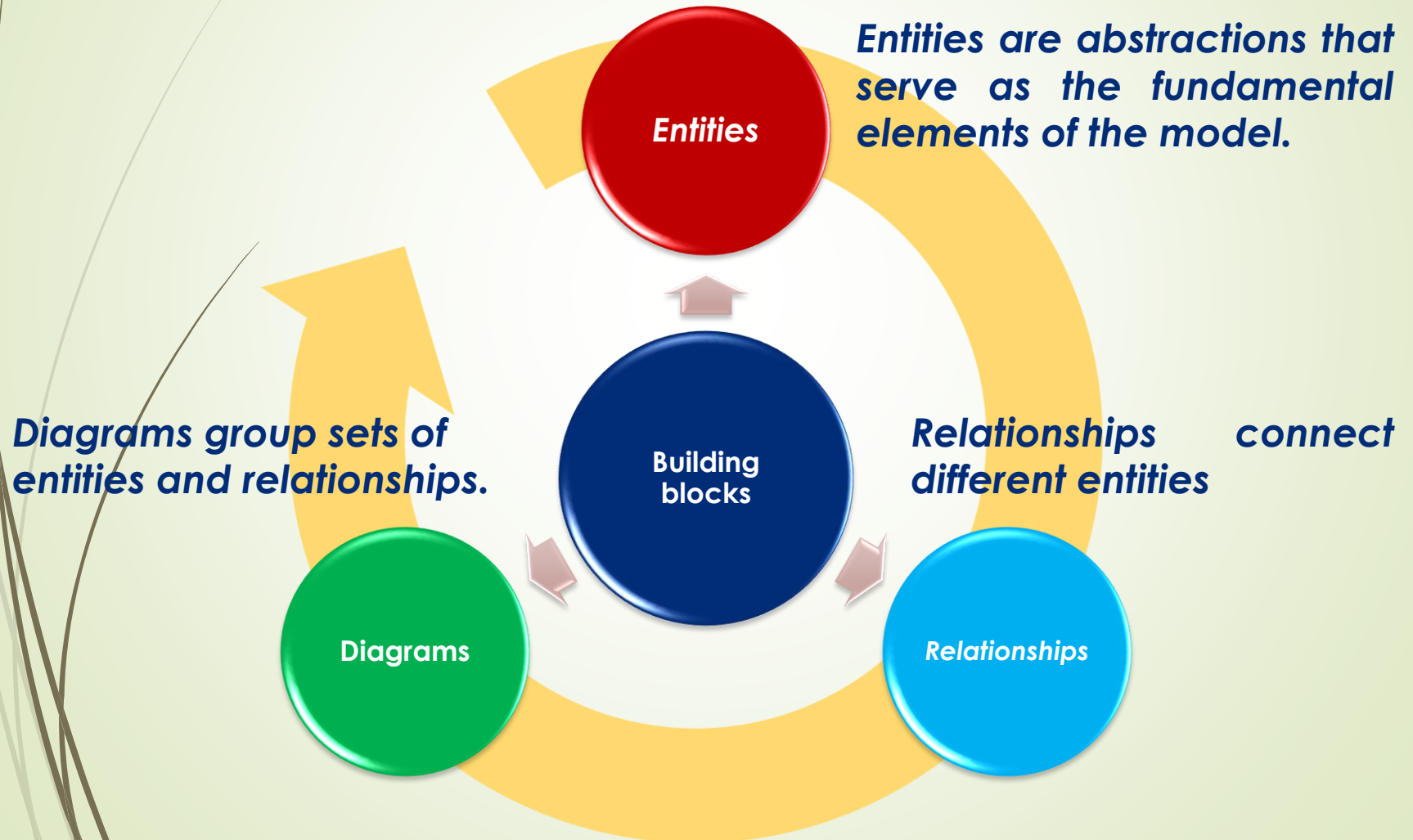
UML is a notation (primarily graphical) used to describe designs.

UML Conceptual Model

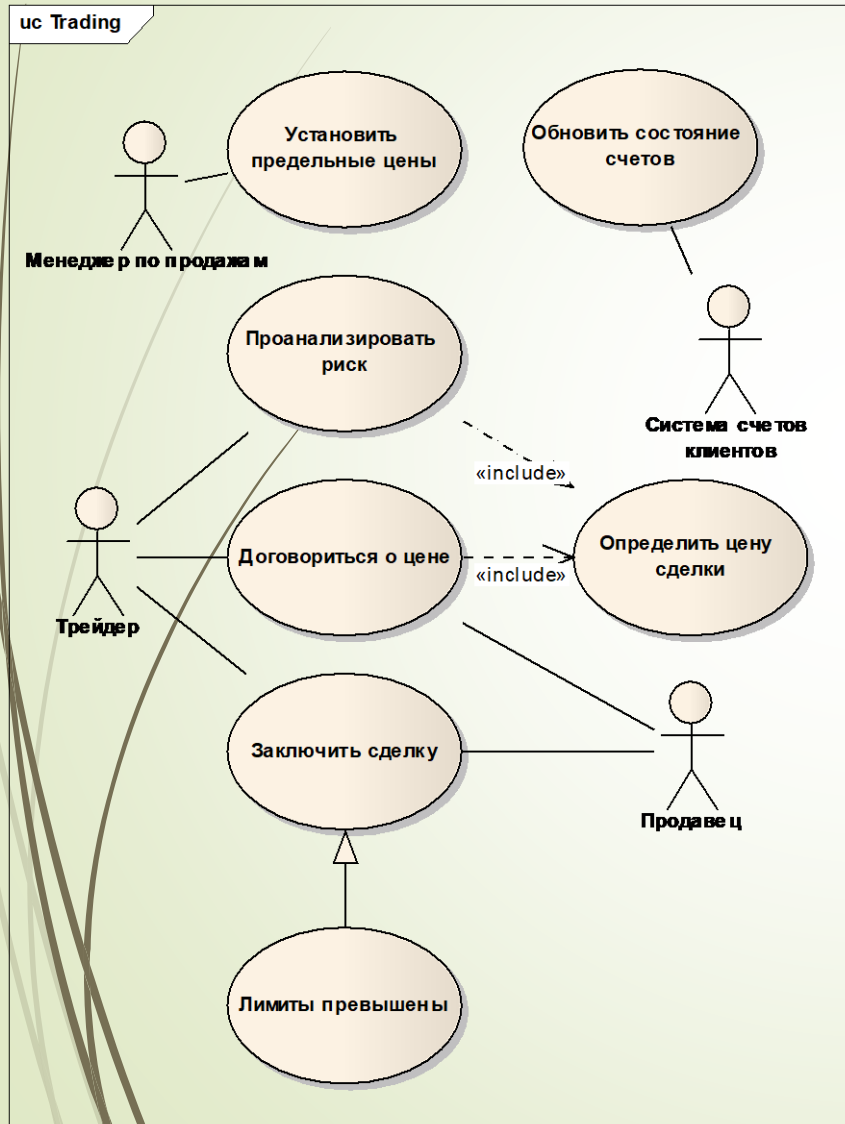
The UML conceptual model comprises three components:

- ✓ Language building blocks
- ✓ Rules for combining building blocks
- ✓ Common language mechanisms

What is UML?



UML. Types of diagrams



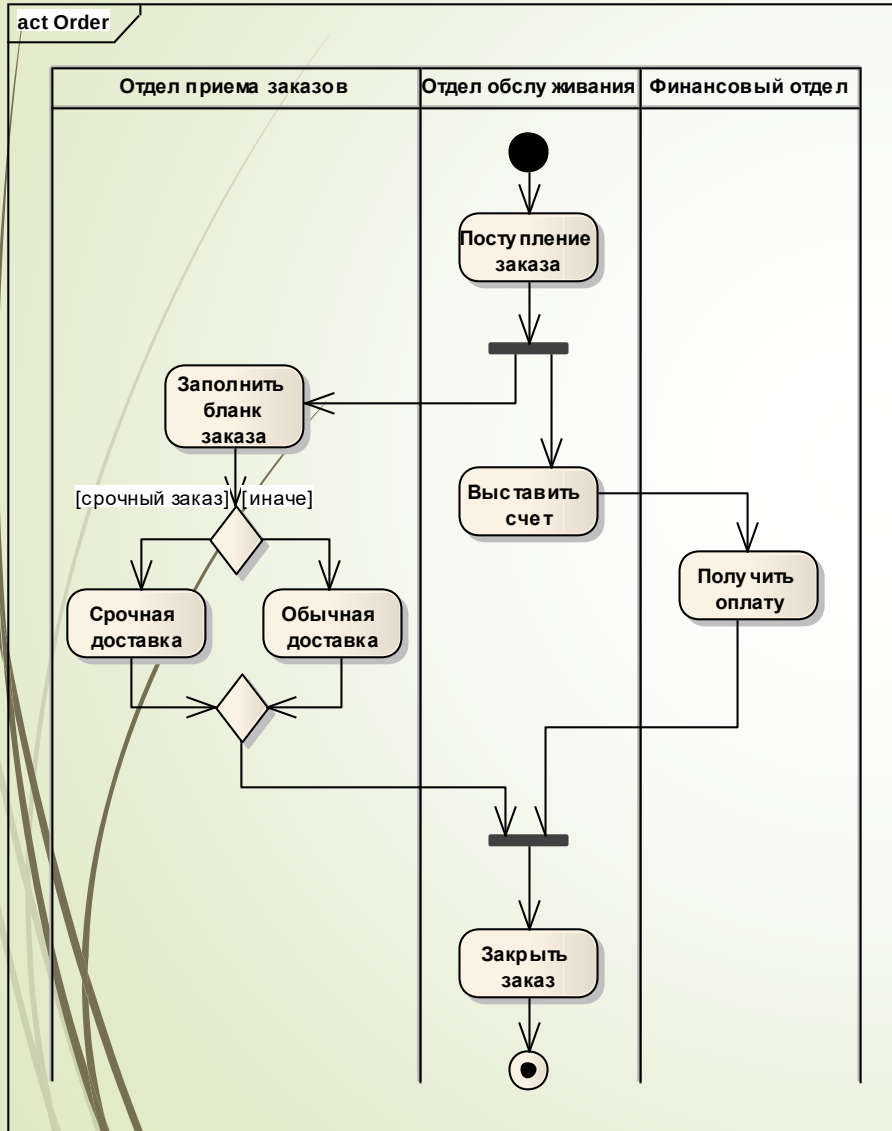
A scenario is a set of steps describing the interaction between a user and the system.

A use case diagram describes a set of scenarios united by a common user goal.

Analyst's perspective:

Use case diagrams serve to represent the system context or functional requirements of the system.

UML. Types of diagrams



An activity is a time-extended state of a system in which something is being performed.

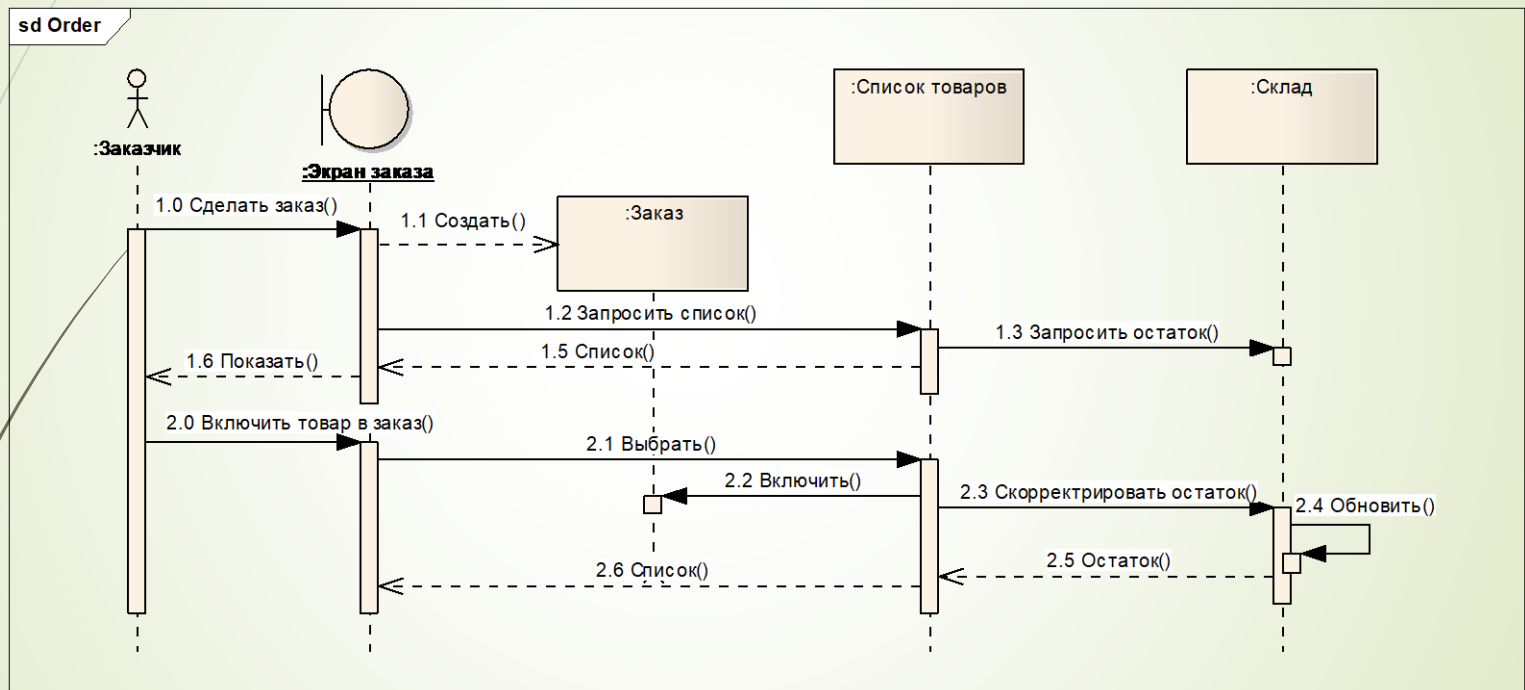
An activity diagram represents the flow of transitions from one activity to another.

Analyst's perspective:

Activity diagrams serve to represent the workflow within a business process.

UML. Types of diagrams

A **sequence diagram** describes the interaction of a set of objects, emphasizing the temporal ordering of messages.

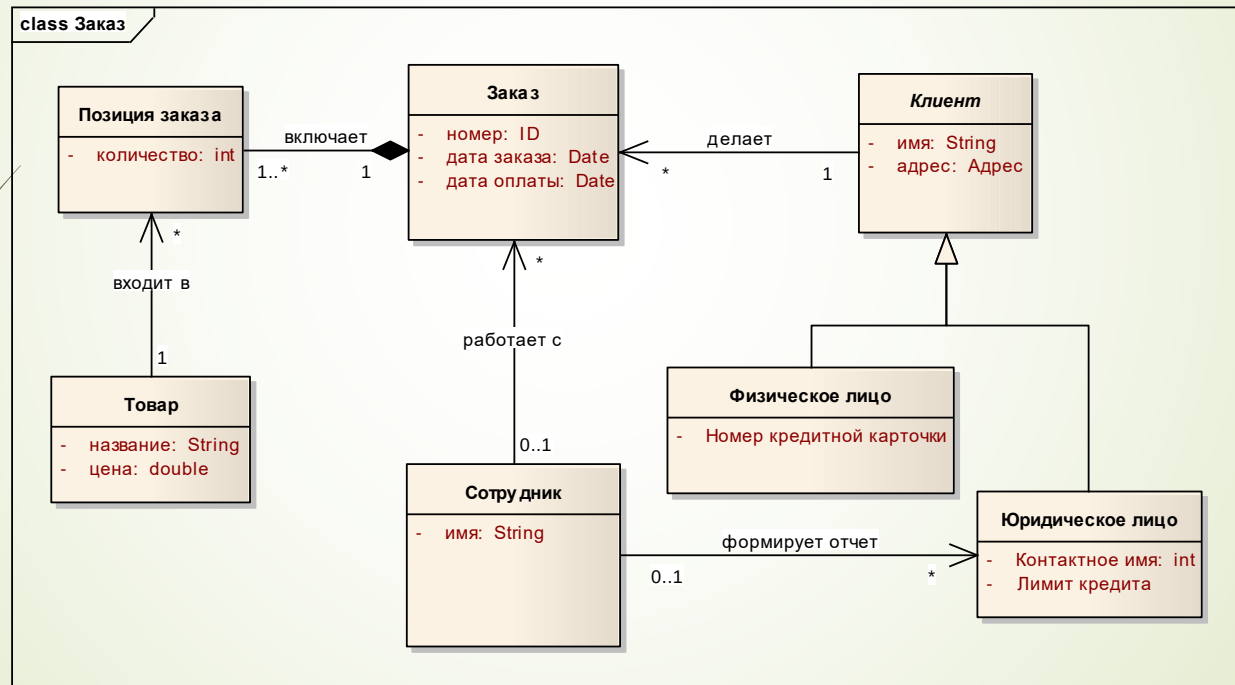


Analyst's Perspective

Sequence diagrams are used to describe the interaction between an actor and the system within the scope of a single use case.

UML. Types of diagrams

A **class diagram** describes the system's object types and the static relationships that exist between them.

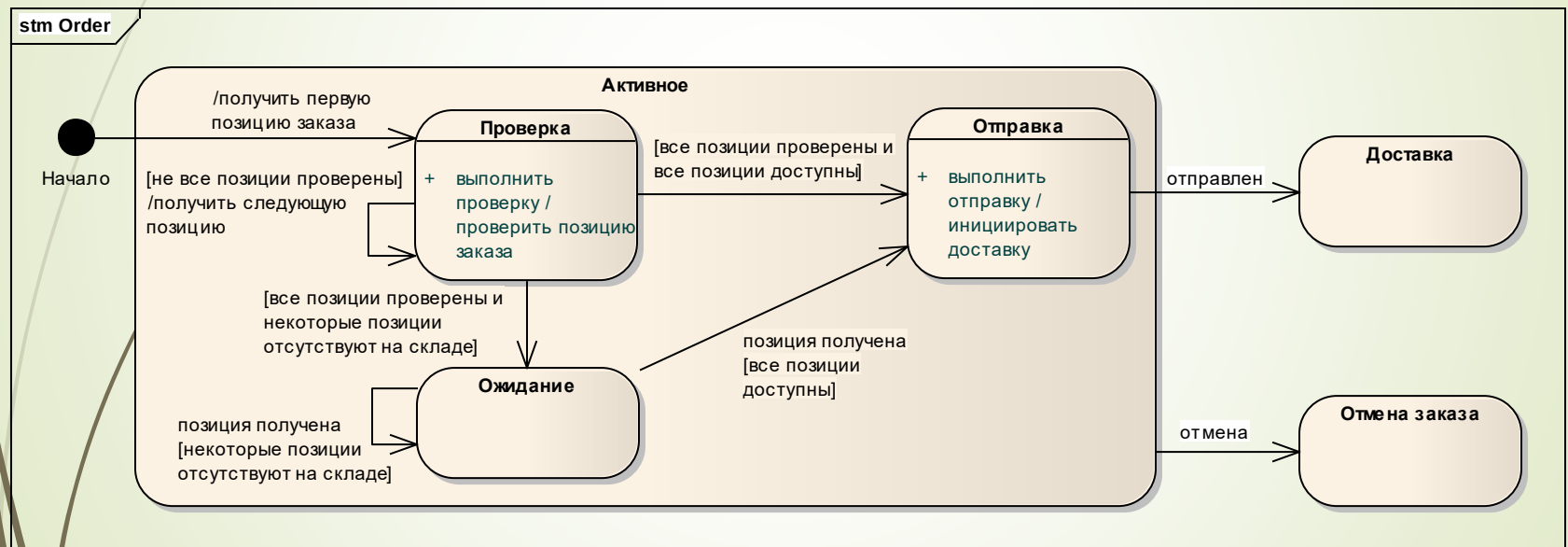


Analyst's perspective

Class diagrams serve to represent concepts of the domain under study.

UML. Types of diagrams

A **state diagram** describes all possible states in which a specific object can exist, as well as changes in the object's state.



Analyst's perspective

State diagrams are used to describe the behavior of a single object across several different use cases.

UML. Model development sequence

Model	Purpose
Use Case Diagram	Description of the system's functional purpose
Activity Diagram	Description of workflows in business processes
Sequence Diagram	System behavior description
Class Diagram	Description of the subject area
State Diagram	Description of events that change the states of objects